

ASSIGNMENT #03

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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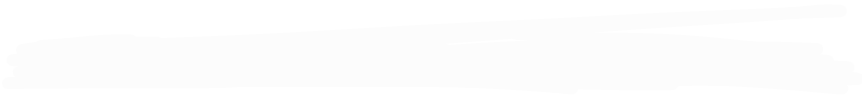
Assignment #03

CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS

Spring 2026

Professor: Dr. Michael Caracotsios

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Abstract

This report presents solutions for CHE 301 Assignment #03, consisting of Problem A and Problem B. In Problem A, thermodynamic properties were evaluated at 10°C and 830 bar using density-based relationships and Akima interpolation (FAKIMA) to obtain needed derivatives and integrands. Properties computed include β , ζ , enthalpy and entropy changes by integration, C_p and C_v , the Joule–Thomson coefficient, and the speed of sound. In Problem B, ideal-gas venting through a narrow pipe section was analyzed under reversible adiabatic (isentropic) assumptions to determine tank mass, critical-section temperature and pressure, pipe diameter, and masses during venting.

Contents

1	Introduction	4
2	Background	5
3	Problem A	6
3.1	A.1 Calculation of Beta, K^{-1}	9
3.2	A.2 Calculation of Zeta, K^{-1}	9
3.3	A.3 Calculation of Enthalpy, $\frac{KJ}{Kmol}$	11
3.4	A.4 Calculation of Entropy, $\frac{KJ}{Kmol \cdot K}$	15
3.5	A.5-6 Calculation of C_p and C_v , $\frac{KJ}{Kmol \cdot K}$	17
3.5.1	A.5 Calculation of C_v , $\frac{KJ}{Kmol \cdot K}$	18
3.5.2	A.6 Calculation of C_p , $\frac{KJ}{Kmol \cdot K}$	23
3.6	A.7 Calculation of the Joule-Thomson Coefficient in $\frac{K}{bar}$	24
3.7	A.8 Calculation of Speed of Sound, $\frac{m}{s}$	25
4	Problem B	26
4.1	B.1. Initial mass of ethylene in the tank,kg:	26
4.2	B.2 The Temperature and Pressure at the narrow pipe section:	27
4.3	B.3 The diameter of the narrow section , mm:	31
4.4	B.4 Mass of ethylene in tank when Critical Conditions are reached, kg: . . .	32
4.5	B.5 Mass of ethylene during the venting Process if $P_{gauge}=P_1=2300$ Psig , kg: . . .	33

5	Results Summary	35
6	Conclusion	36
7	References	37

1 Introduction

This assignment applies core chemical engineering thermodynamics tools to compute real-gas properties and to analyze compressible flow under ideal-gas/isentropic assumptions. Problem A focuses on evaluating properties at a specified state (10°C and 830 bar) using tabulated data, interpolation, and thermodynamic identities written in terms of density. Problem B analyzes the venting of ethylene from a tank through a narrow pipe section, including the critical (sonic) condition and associated flow/geometry requirements.

The objective of this report is to present the calculations, intermediate steps, and final results in a clear, typed format while preserving the original workflow used to obtain the reported numerical values.

2 Background

Thermodynamic properties for real fluids can be obtained from tabulated data and evaluated at intermediate states using interpolation. In this report, Akima interpolation (FAKIMA) is used to estimate values and derivatives needed for property calculations.

For Problem A, the analysis uses density-based identities and partial-derivative relations to compute:

- The compressibility factor Z at the target state using interpolation and averaging approaches.
- The thermal expansion coefficient β from $\left(\frac{\partial \rho}{\partial T}\right)_P$.
- The coefficient ζ from $\left(\frac{\partial P}{\partial T}\right)_\rho$ using derivative transformations.
- Enthalpy and entropy changes using integrals written in terms of pressure (and later density) with $V = \frac{1}{\rho}$.
- Heat capacities C_p and C_v using thermodynamic identities involving $\left(\frac{\partial P}{\partial T}\right)_\rho$ and $\left(\frac{\partial P}{\partial \rho}\right)_T$, plus a density-form integral for C_v requiring second derivatives.
- The Joule–Thomson coefficient μ_{JT} and the speed of sound v using the computed properties and required conversions.

For Problem B, the venting process is modeled as ideal-gas behavior with reversible adiabatic (isentropic) relations. Conservation of energy and entropy relations connect tank conditions to the narrow section at critical conditions, and continuity relates mass flow rate to density, velocity, and area to determine the required diameter.

3 Problem A

To begin this problem we would have to find the Z value at 10°C and 830 bar in the table and in order to find it we will use the CHE calculator. We will first copy the table into excel and then use the function fakima as follows:

- We will find Z at a constant temperature first

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 1, X_{\text{needed}}),$$

where 1 means value/interpolation at, the $X_i : X_f$ are the temperature values we have, $Y_i : Y_f$ are the values of Z at each pressure given at 10°C, and the X_{needed} is the value that we need to interpolate at which in this case is the 10°C

So at 800 atm we get,

$$\text{fakima}(0 : 100, 1.798 : 1.625, 1, 10)$$

and then we will do that for every pressure from 0 atm to 1000 atm and then we will use this table to find at 830 bar = 819.1463 atm which give us Z= 1.785362 which is the Z at constant temperature

Which gives us a $Z = 1.786871767$ We will do the same at constant temperature and get a Z value $Z = 1.792799148$

Now that we have all possible values of Z we can take the average to give us

$$Z = 1.787590612$$

New Table with new values						
	0	10	25	50	75	100
0	1	1	1	1	1	1
1	1	1	1	1	1	1
3	0.999	0.99888	0.999	1	1	1.001
5	0.998	0.9984	0.999	1	1.001	1.001
10	0.996	0.9968	0.998	1	1.001	1.002
30	0.959	0.979082909	0.996	1.001	1.005	1.007
50	0.958	0.977978182	0.996	1.004	1.009	1.014
100	0.984	0.992746182	1.004	1.018	1.027	1.033
200	1.036	1.045146182	1.057	1.072	1.082	1.089
300	1.132	1.138464	1.146	1.154	1.16	1.163
400	1.256	1.255096	1.254	1.253	1.25	1.247
600	1.524	1.51174	1.495	1.471	1.45	1.432
800	1.798	1.760754323	1.723	1.697	1.658	1.625
819.146311	1.823950415	1.787590612	1.746072248	1.718635332	1.677995054	1.643559081
1000	2.067	2.033573913	1.988	1.923	1.867	1.819

Figure 3: Full table with correct Z value at 830 and 10°C

Using our Z values we can make a density table where $\rho = P/RT$ to get the following table of densities in Kmol/m^3 :

Density, Kmol/m^3								
P(bar)	T(K)							
	273.15	283.15	298.15	323.15	348.15	373.15	To calculate density	
	0	0	0	0	0	0	$\rho = P/RTZ$	
1.01325	0.044618	0.043042	0.040876	0.037714	0.035006	0.032661	conversion factors	
3.03975	0.133987	0.12927	0.122752	0.113142	0.105017	0.097884	1 bar*m ³ ---->KJ	
5.06625	0.223535	0.215554	0.204586	0.18857	0.174854	0.163139	R, KJ/Kmol*m ³	
10.1325	0.447967	0.431799	0.409582	0.37714	0.349708	0.325953	$\rho = P*100/RTZ$	
30.3975	1.395751	1.318839	1.231214	1.130289	1.04495	0.973005		
50.6625	2.32868	2.200548	2.052024	1.878187	1.734679	1.610479		
101.325	4.5343	4.335626	4.071347	3.704714	3.408551	3.161716		
202.65	8.61342	8.236505	7.734403	7.036191	6.470576	5.99826		
303.975	11.82443	11.34206	10.70061	9.804329	9.053228	8.424899		
405.3	14.2094	13.71744	13.0387	12.03958	11.20186	10.47651		
607.95	17.56595	17.083	16.40521	15.383	14.48516	13.68458		
810.6	19.85206	19.55606	18.97914	17.77913	16.89062	16.07903		
830	20.03798	19.72348	19.17658	17.97547	17.08878	16.27793		
1013.25	21.58564	21.16557	20.56153	19.61206	18.74977	17.95521		

Figure 4: Density Table

Once we have that we can start calculating the unknown variables.

3.1 A.1 Calculation of Beta, K⁻¹

$$\beta = \frac{-1}{\rho} \times \left(\frac{d\rho}{dT} \right)_P$$

where,

$$\left(\frac{d\rho}{dT} \right)_P$$

is the slope and to find it we will use the fakima again as follows

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 3, X_{\text{needed}}),$$

where 3 is the first derivative, the X values are the temperatures, the Y values are the densities at 830 bar and the X_needed is 10°C. For example

$$= \text{fakima}(273.15 : 373.15, 20.038 : 16.278, 3, 283.15)$$

To give us a slope or $\left(\frac{d\rho}{dT} \right)_P = -0.03296258$.

Now we plug everything in to find Beta,

$$\beta = -1/(19.72348) \times -0.03296258$$

$$\boxed{= 0.00167124\text{K}^{-1}}$$

3.2 A.2 Calculation of Zeta, K⁻¹

$$\zeta = \frac{1}{P} \times \frac{dP}{dT}_V$$

Since we don't have a volume we know that $V = \frac{1}{\rho}$ which means

$$\zeta = \frac{1}{P} \left(\frac{\partial P}{\partial T} \right)_{\rho}$$

To find $\left(\frac{dP}{dT} \right)_{\rho} \rightarrow d\rho = \left(\frac{d\rho}{dT} \right)_P dT + \left(\frac{d\rho}{dP} \right)_T dP$ now solve for

$$\left(\frac{\partial P}{\partial T} \right)_{\rho} = - \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T}, \text{ so,}$$

$$\zeta = \frac{1}{P} \left(- \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T} \right)$$

We already found $\left(\frac{d\rho}{dT} \right)_P = -0.03296258$ now to find $\left(\frac{d\rho}{dP} \right)_T$ we will still use fakima in the CHE calculator as follows,

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 3, X_{\text{needed}}),$$

where 3 is the first derivative, the X values are the pressures, the Y values are the densities at 10°C and the X_needed is 830 bar. For example

$$= \text{fakima}(0 : 1013.25, 0 : 21.16557, 3, 830)$$

To give us a slope or $\left(\frac{d\rho}{dP} \right)_T = 0.008002965$

Now plug everything into the equation to get Zeta,

$$\zeta = 1/(830) * (-0.008002965 / -0.03296258)$$

$$\boxed{= 0.00492405 \text{K}^{-1}}$$

Finding Beta			
Beta= $-1/p * dp/dT)_P$			
finding $dp/dT)_P$		-0.03296258	
Beta=		0.001671236	
Finding Zeta			
$zeta=1/P * dP/dT)_V$			
which also means			
$Zeta=1/P * dP/dT)_P$			
Fidning $dP/dT)_P$			
$dp =$		$dp/dT)_PdT + dp/dP)_TdP$	
density is constant so $dp=0$			
solve for $dP/dT)_P$			
$dP/dT)_P = (-dp/dT)_P / (dp/dP)_T$			
$zeta=1/P * (-dp/dT)_P / (dp/dP)_T$			
Now find each derivative			
$dp/dT)_P$		-0.03296258	
$dp/dP)_T$		0.008002965	
zeta=		0.004962405	

Figure 5: Beta and Zeta Excel Calculations

3.3 A.3 Calculation of Enthalpy, $\frac{KJ}{Kmol}$

We know that the formula for enthalpy is

$$dH = C_p dT + (1 - \beta T) V dP$$

We also know that T is constant and $V = \frac{1}{\rho}$ so that gives us

$$dH = (1 - \beta) \frac{1}{\rho} dP$$

At 0 bar we know that it is ideal gas so we can use the HIG function in the CHE calculator to find the enthalpy at 0 bar and 10°C to get $-0.43776 \frac{KJ}{Kmol}$

So now we will need to find β at each pressure and in order to do that we will need to find the slope $\left(\frac{d\rho}{dT}\right)_P$ for each pressure at 10°C to get the following table

	T(K)	283.15
Pressure (bar)	Slope or $dp/dT)_P$	
0	0	
1.01325	-0.00015	
3.03975	-0.00046	
5.06625	-0.00077	
10.1325	-0.00156	
30.3975	-0.00676	
50.6625	-0.01137	
101.325	-0.0189	
202.65	-0.03587	
303.975	-0.04582	
405.3	-0.04751	
607.95	-0.04699	
810.6	-0.03387	
830	-0.03296	
1013.25	-0.04126	

Figure 6: $\left(\frac{d\rho}{dT}\right)_P$ Calculation

Now that we have the slope at each pressure we can find β at each pressure and we calculate that just like we calculated the Beta in problem 1 and once we do that we will get

	T(K)	283.15
Pressure (bar)	Beta	
0	0	
1.01325	-0.00015	
3.03975	-0.00046	
5.06625	-0.00077	
10.1325	-0.00156	
30.3975	-0.00676	
50.6625	-0.01137	
101.325	-0.0189	
202.65	-0.03587	
303.975	-0.04582	
405.3	-0.04751	
607.95	-0.04699	
810.6	-0.03387	
830	-0.03296	
1013.25	-0.04126	

Figure 7: Beta Calculation

Now we can do the integrand at each pressure which is $(1 - \beta) * 1/\rho$ at 10°C and we will get the following table

Now we will find the integrand for each pressure at 10°C		
3	Pressure (bar)	Integrand
9	0	0
0	1.01325	0.00788
1	3.03975	0.01121
2	5.06625	-0.05104
3	10.1325	-0.05182
4	30.3975	-0.34143
5	50.6625	-0.2102
6	101.325	-0.05397
7	202.65	-0.0283
8	303.975	-0.01268
9	405.3	0.00141
0	607.95	0.01295
1	810.6	0.02606
2	830	0.02671
3	1013.25	0.02117

Figure 8: Enthalpy Integrand Calculation

Now that we have the integrand at each pressure we will be using the fakima function in excel to find the enthalpy and we do that as follows

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 5, \text{lower limit}, \text{higher limit}),$$

where 5 is the integral, the X values are the pressures, the Y values are the integrand at each pressure at 10°C, the lower limit is 0 and the higher limit is 830 bar. For example

$$= \text{fakima}(0 : 1013.25, 0 : 0.026709, 5, 0, 830)$$

And that will give us

$$H = -16.86897077 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol}}$$

to convert to KJ we will multiply by

$$100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$$

to get

$$H = -1686.897077 \frac{\text{KJ}}{\text{Kmol}}$$

and then now we can convert to $\frac{\text{KJ}}{\text{mol}}$ to get

$$H = -1.686897077 \frac{\text{KJ}}{\text{mol}}$$

and now to get the total enthalpy we will add the enthalpy at 0 bar and the enthalpy that we integrated to give us:

$$H_{\text{tot}} = -2.124659675 \frac{\text{KJ}}{\text{mol}}$$

5	Now we can find the integral, bar*m^3/kmol	
6		-16.86897077
7	Convert to KJ/Kmol	
8		-1686.897077
9	Convert to KJ/mol	
0		-1.686897077
1		
2	total enthalpy, HIG at 0 bar and the integral, KJ/mol	
3		-2.124659675
4		
5		

Figure 9: Enthalpy Calculation

3.4 A.4 Calculation of Entropy, $\frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$

We know the formula for entropy is

$$ds = C_p \left(\frac{dT}{T} \right) - \beta V dP$$

We know that T is constant and $V = \frac{1}{\rho}$ so the entropy formula becomes

$$ds = -\frac{\beta}{\rho} dP$$

Since we already calculated the Beta at each pressure and at constant temperature already for the previous question we now can do the integrand at each pressure which is just

$$-\frac{\beta}{\rho}$$

and ρ is at constant temperature (10°C), once we do that we get the following table

Pressure (bar)	Integrand
0	0
1.01325	-0.081933936
3.03975	-0.027620856
5.06625	-0.016750947
10.1325	-0.011861931
30.3975	-0.003916546
50.6625	-0.001980484
101.325	-0.001004446
202.65	-0.00049046
303.975	-0.000305358
405.3	-0.000200525
607.95	-0.000101377
810.6	-8.54587E-05
830	-9.88303E-05
1013.25	0

Figure 10: Entropy Integrand Calculation

Now with integrand at each pressure we can take the integral the same way we did for the enthalpy using fakima function in the CHE calculator and once we do that we get

$$S = -0.741618391 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol} \cdot \text{K}}$$

to get it in $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$ we multiply by

$$100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$$

to get

$$S = -74.1618391 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

also we know that the entropy at 0 bar should be ideal gas but since the function in the

calculator doesn't provide an entropy at 0 bar we will take it at 1 bar to get

$$SIG = -1.397044779 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and to get the total entropy we will add the SIG and the S calculated by the integral to get

$$S_{\text{tot}} = -75.55888391 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

Now we can find the integral, bar*m^3/kmol*K	
-0.741618391	
Convert to KJ/Kmol*K	
-74.16183914	
at P=0 bar at 10 C it is ideal gas so SIG, KJ/Kmol K=	
-1.397044779	
Total Entropy	
-75.55888391	

Figure 11: Entropy Calculation

3.5 A.5-6 Calculation of C_p and C_v , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

We know that

$$C_p - C_v = \left(\frac{T}{\rho^2} \right) \left(\frac{\left(\left(\frac{\partial P}{\partial T} \right)_\rho \right)^2}{\left(\frac{\partial P}{\partial \rho} \right)_T} \right)$$

And

$$\left(\frac{\partial P}{\partial T}\right)_\rho = -\frac{\left(\frac{\partial \rho}{\partial T}\right)_P}{\left(\frac{\partial \rho}{\partial P}\right)_T}$$

We have previously calculated $\left(\frac{\partial \rho}{\partial T}\right)_P = -0.03296258$ and $\left(\frac{\partial \rho}{\partial P}\right)_T = 0.008002965$ so therefore

$$\left(\frac{\partial P}{\partial T}\right)_\rho = -\frac{\left(\frac{\partial \rho}{\partial T}\right)_P}{\left(\frac{\partial \rho}{\partial P}\right)_T} = -\left(\frac{-0.03296258}{0.008002965}\right)$$

to give us

$$\left(\frac{\partial P}{\partial T}\right)_\rho = 4.118795909$$

Then we can calculate $C_p - C_v$ such that

$$C_p - C_v = \left(\frac{283.15}{(19.72347533)^2}\right) \left(\frac{(4.118795909)^2}{0.008002965}\right)$$

to give us

$$C_p - C_v = 0.098819153 \frac{\text{m}^3 \cdot \text{bar}}{\text{Kmol} \cdot \text{K}}$$

and now we convert it to $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$ to give us

$$C_p - C_v = 9.881915289 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

3.5.1 A.5 Calculation of C_v , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

$$dU = C_v dT + (\zeta T - 1) P dV \quad \zeta = \frac{1}{P} \left(\frac{\partial P}{\partial T}\right)_\rho$$

Using the Maxwell equations

$$\left(\frac{\partial C_v}{\partial V}\right)_T = -\frac{\partial}{\partial T} [(\zeta T - 1) P]_V$$

$$\frac{\partial}{\partial T} \left[\left(1 + \frac{\partial P}{\partial T} \right) T \right]_V - \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right]_V$$

$$T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho + \left(\frac{\partial P}{\partial T} \right)_\rho - \left(\frac{\partial P}{\partial T} \right)_\rho \Rightarrow \boxed{\left(\frac{\partial C_v}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho}$$

$$\int dC_v = \int T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho dV \Rightarrow C_v = C_v^{IG} + \int_\infty^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho dV$$

We will need to evaluate the integral in terms of density such as

$$C_{vIG} + \int_0^\rho \left(T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho \right) \left(\frac{d\rho}{-\rho^2} \right)$$

and we can get the CvIG by using the CHE calculator to get

$$C_{vIG} = 20.87745872$$

and now we will evaluate

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \frac{\rho_{TT}\rho_P^2 - 2\rho_T\rho_{TP}\rho_P + \rho_T^2\rho_{PP}}{\rho_P^3}$$

Definitions:

$$\rho_T = \left(\frac{\partial \rho}{\partial T} \right)_P \quad \rho_P = \left(\frac{\partial \rho}{\partial P} \right)_T$$

$$\rho_{TT} = \left(\frac{\partial^2 \rho}{\partial T^2} \right)_P \quad \rho_{PP} = \left(\frac{\partial^2 \rho}{\partial P^2} \right)_T$$

$$\rho_{TP} = \frac{\partial^2 \rho}{\partial T \partial P} = \frac{\partial}{\partial P} \left(\frac{\partial \rho}{\partial T} \right)$$

To calculate each derivative we will use the function fakima just the same way we have used it to get previous slopes or derivatives all of those calculations were done in excel using the fakima function so I will be putting tables to show the numbers that I have got for each derivative at each point.

dp/dT)_P						
	T(K)					
Pressure (bar)	273.15	283.15	298.15	323.15	348.15	373.15
0	0	0	0	0	0	0
1.01325	-0.000164182	-0.000151959	-0.000136839	-0.000116469	-0.000100258	-8.65541E-05
3.03975	-0.000490192	-0.000455881	-0.000415275	-0.0003512	-0.000301208	-0.000265535
5.06625	-0.000831561	-0.000769645	-0.000693053	-0.000591812	-0.000505822	-0.00042857
10.1325	-0.001684583	-0.00155911	-0.001407104	-0.001186445	-0.001012434	-0.000876678
30.3975	-0.008616006	-0.00675506	-0.00449194	-0.003556301	-0.003125438	-0.002609909
50.6625	-0.01426904	-0.011366479	-0.007820593	-0.005992168	-0.005268401	-0.004581792
101.325	-0.020991864	-0.01889531	-0.01630805	-0.012975499	-0.010685854	-0.00888683
202.65	-0.03980046	-0.035869158	-0.031017156	-0.024758261	-0.020434029	-0.017026655
303.975	-0.050973553	-0.045818237	-0.039409626	-0.032456032	-0.027383289	-0.022677739
405.3	-0.051169254	-0.047508779	-0.043244508	-0.03647601	-0.030858905	-0.026766579
607.95	-0.049849501	-0.046989922	-0.043533249	-0.038277089	-0.033730332	-0.030078584
810.6	-0.025170929	-0.03386758	-0.042425374	-0.038579007	-0.033073059	-0.030925507
830	-0.028945012	-0.032962578	-0.03976007	-0.038077752	-0.033023326	-0.030916777
1013.25	-0.042874793	-0.041257568	-0.039508136	-0.036381129	-0.032966894	-0.030427721

Figure 12: $\left(\frac{\partial \rho}{\partial T} \right)_P$

(d^2p/dT^2)_P						
	T(K)					
Pressure (bar)	273.15	283.15	298.15	323.15	348.15	373.15
0	0	0	0	0	0	0
1.01325	1.51923E-06	1.02241E-06	8.5331E-07	6.57316E-07	4.50872E-07	6.45475E-07
3.03975	4.26486E-06	3.11556E-06	2.28639E-06	2.29328E-06	9.52041E-07	1.90179E-06
5.06625	7.6956E-06	5.17897E-06	4.47752E-06	3.48494E-06	2.75561E-06	3.42452E-06
10.1325	1.55953E-05	1.0923E-05	8.60431E-06	7.48434E-06	4.07458E-06	6.78584E-06
30.3975	0.000182683	6.36168E-05	3.43332E-05	-2.16796E-07	1.81907E-05	2.30516E-05
50.6625	0.000292975	0.000113165	6.18283E-05	2.5431E-06	1.71756E-05	3.77532E-05
101.325	0.000255337	0.000165706	0.000127653	8.77824E-05	5.10684E-05	9.28536E-05
202.65	0.000479131	0.000311329	0.000240576	0.000166134	9.73424E-05	0.000175248
303.975	0.000610928	0.000367374	0.000297754	0.000173058	0.000163584	0.00021286
405.3	0.000451864	0.000335159	0.000245671	0.000262765	0.000115396	0.00021199
607.95	0.000360789	0.000260689	0.000214245	0.000203553	0.000117497	0.000174643
810.6	-0.000918708	-0.000696377	-0.00164567	0.000288785	-2.55847E-05	0.000197389
830	-0.000699485	-0.000492639	-0.00212281	0.00022209	-2.70155E-05	0.000195539
1013.25	0.000197566	0.000161875	0.000116865	0.000180348	8.11572E-05	0.000121977

Figure 13: $\left(\frac{\partial^2 \rho}{\partial T^2} \right)_P$

dp/dP)_T							
P(bar)	T(K)						
		273.15	283.15	298.15	323.15	348.15	373.15
0	0.044001006	0.042443193	0.040311504	0.03722081	0.034548053	0.032257582	
1.01325	0.04406236	0.042530608	0.04038722	0.03722081	0.034548053	0.032197205	
3.03975	0.044133182	0.042561356	0.04039372	0.03722081	0.034548053	0.032192037	
5.06625	0.044192392	0.04258025	0.040398273	0.03722081	0.03447705	0.032196601	
10.1325	0.044624206	0.04299603	0.040517239	0.03722081	0.034480594	0.032112118	
30.3975	0.046405273	0.043655346	0.040539851	0.037149128	0.034260854	0.03183572	
50.6625	0.045579386	0.043417012	0.040496815	0.03684146	0.033947868	0.031329083	
101.325	0.042794944	0.041602997	0.039540277	0.035630679	0.03254934	0.030168696	
202.65	0.037800896	0.035862343	0.03357801	0.030786725	0.028341728	0.026316079	
303.975	0.02719621	0.026797552	0.026078314	0.024690277	0.023407023	0.022179763	
405.3	0.019305533	0.019200622	0.019112564	0.019115671	0.018717655	0.018128988	
607.95	0.012315338	0.013715621	0.013799865	0.012920142	0.012947111	0.01286488	
810.6	0.009883869	0.009156041	0.011189627	0.010465867	0.010561576	0.010589064	
830	0.008901871	0.008002965	0.008891705	0.009419887	0.009535481	0.009607381	
1013.25	0.007876953	0.007489403	0.006248006	0.008336223	0.008489003	0.008602899	

Figure 14: $\left(\frac{\partial \rho}{\partial P}\right)_T$

d^2p/dP^2)_T						
Pressure (bar)	T(K)					
	273.15	283.15	298.15	323.15	348.15	373.15
0	7.46544E-05	3.89868E-05	2.98906E-05	2.28718E-18	-1.55689E-17	-2.38351E-05
1.01325	4.20854E-05	2.81343E-05	3.8421E-05	3.07966E-18	-3.15657E-18	-3.06529E-05
3.03975	0.000105522	2.96209E-05	-3.86839E-05	-2.73859E-18	-0.000185391	2.27131E-05
5.06625	-4.3609E-05	-4.17655E-05	2.9809E-05	1.44453E-19	4.18164E-05	-3.72867E-05
10.1325	0.0004594	0.00016468	5.80734E-06	-9.43938E-06	-2.95658E-05	-2.67902E-05
30.3975	-2.7685E-05	-1.98209E-05	-6.42488E-06	-4.16615E-05	-3.58397E-05	-6.21483E-05
50.6625	-0.000132131	-7.92517E-05	-3.78555E-05	-4.55985E-05	-5.23497E-05	-3.82844E-05
101.325	-5.16643E-05	-7.05113E-05	-8.29785E-05	-6.7322E-05	-5.48905E-05	-5.27004E-05
202.65	-0.000152527	-0.000129758	-0.000106822	-8.49848E-05	-7.1534E-05	-5.85191E-05
303.975	-6.0883E-05	-4.86794E-05	-4.03394E-05	-4.57059E-05	-3.78116E-05	-3.44434E-05
405.3	-1.22036E-05	-2.26361E-05	-2.15903E-05	-1.63437E-05	-1.75365E-05	-1.60981E-05
607.95	-6.62483E-06	2.31246E-07	-6.76375E-06	-8.23237E-06	-8.34707E-06	-8.60346E-06
810.6	8.18933E-06	-4.38459E-05	-7.62327E-05	9.56776E-07	-1.65851E-06	-2.7404E-06
830	-3.75197E-06	1.23752E-06	-1.48239E-05	-4.18282E-06	-4.01393E-06	-3.91692E-06
1013.25	-7.43404E-06	-6.84255E-06	-1.40296E-05	-7.64435E-06	-7.40738E-06	-7.04605E-06

Figure 15: $\left(\frac{\partial^2 \rho}{\partial P^2}\right)_T$

d/dP(dp/dT)						
P(bar)	T(K)					
	273.15	283.15	298.15	323.15	348.15	373.15
0	-0.000162616	-0.000149971	-0.000133875	-0.000114504	-9.88405E-05	-8.39732E-05
1.01325	-0.000161881	-0.000149972	-0.000133875	-0.000114504	-9.88405E-05	-8.39732E-05
3.03975	-0.000167971	-0.000149985	-0.000133875	-0.000114504	-9.88405E-05	-8.39724E-05
5.06625	-0.000168449	-0.000154876	-0.000133875	-0.000114504	-9.88405E-05	-8.39733E-05
10.1325	-0.00016859	-0.000159191	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
30.3975	-0.000307795	-0.000240241	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
50.6625	-0.000199425	-0.000179871	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
101.325	-0.000167629	-0.000158677	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
202.65	-0.000160894	-0.000154462	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
303.975	-1.28478E-05	-3.43933E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
405.3	2.15971E-06	-5.76883E-06	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
607.95	9.50943E-06	3.46067E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
810.6	-3.41763E-05	5.74475E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
830	-0.000108321	3.15245E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
1013.25	-1.67526E-05	-9.12237E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05

Figure 16: $\frac{\partial^2 \rho}{\partial T \partial P}$

Now that we have all the numbers needed to solve for the integral of Cv we will now do the integrand which is

$$= \left(\frac{\partial^2 P}{\partial T^2} \right)_{\rho} \left(\frac{-1}{\rho^2} \right) d\rho$$

Once we plug everything in at 10°C and the densities at 10°C we get the following integrand table

Since we are using the density as the integral limits we will need to use the densities at 10 C because it is Cv(T,p)			
Densities, Kmol/m ³	Integrand= T*(d ² P/dT ²) _p *(-1/p ²))dp		
0	0		
0.04304176	0.00062077		
0.129270064	0.00013226		
0.215553688	0.000219119		
0.431799363	5.05849E-05		
1.318839091	0.000147575		
2.200548093	-6.44672E-05		
4.335625889	-8.98241E-06		
8.236505327	5.23907E-05		
11.34206101	-3.11707E-05		
13.71744009	-4.65063E-05		
17.08300123	-0.000125051		
19.556055	0.000248825		
19.72347533	6.80822E-05		
21.16557465	0.000313205		

Figure 17: Cv Integrand Calculation

Once we have that we can now integrate using fakima function where the densities from 0 bar to 830 bar at 10°C are the limits or the X values in fakima and the Y values are the integrand that we just calculated and multiply it by the temperature in Kelvin= 283.15 K and multiply by 100 KJ to get rid of the bar · m³ Once we do that we get

$$C_v = 2.44761456 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and to get the total C_v we will add the C_{vIG} to the integral that we evaluated to get

$$C_v = 20.87745872 + 2.44761456$$

to give us

$$C_v = 23.32511018 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

3.5.2 A.6 Calculation of $C_p, \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

To find C_p we will use

$$C_p - C_v = 9.881915289 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and we just calculated

$$C_v = 23.32511018 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

so now we just solve for

$$C_p = 9.881915289 + 23.32511018$$

to give us

$$C_p = 33.20702547 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

Integral*100 to convert to KJ/Kmol*K	
2.447651456	
Cv=	23.32511018
Cp=Cv+BetaTZetaP/p	33.20702547

Figure 18: C_p and C_v calculation

3.6 A.7 Calculation of the Joule-Thomson Coefficient in $\frac{\text{K}}{\text{bar}}$

We know that the formula for the Joule Thomson Coefficient is

$$\mu_{JT} = \frac{1}{C_p} (\beta T - 1) \frac{1}{\rho}$$

and multiply by $100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$ to get it in

$$\frac{\text{K}}{\text{bar}}$$

$$\mu_{JT} = \frac{1}{33.207} \left(\frac{(0.00161)(283.15) - 1}{19.723} \right) \times 100$$

To give

$$\mu_{JT} = -0.080431055 \frac{\text{K}}{\text{bar}}$$

3.7 A.8 Calculation of Speed of Sound, $\frac{\text{m}}{\text{s}}$

we know that the formula for the speed of sound is

$$v^2 = \frac{1}{M_w} \gamma \left(\frac{\partial P}{\partial \rho} \right)_T$$

we know that:

$$M_w = 28.013 \frac{\text{kg}}{\text{Kmol}}$$

$$\left(\frac{\partial P}{\partial \rho} \right)_T \text{ was previously calculated which is } 124.9536948 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol}}$$

$$\gamma = \frac{C_p}{C_v} = \frac{33.207}{23.325}, \gamma = 1.423659962$$

we will need to Convert to get units of $\frac{\text{m}}{\text{s}}$ so we will use

$$\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \quad \text{and} \quad \frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}}$$

This will give us

$$v = \sqrt{\left(\frac{1}{28.013} \right) (1.42)(124.951) \left(\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \right) \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}} \right)}$$

Which will give us,

$$v = 796.8890813 \frac{\text{m}}{\text{s}}$$

4 Problem B

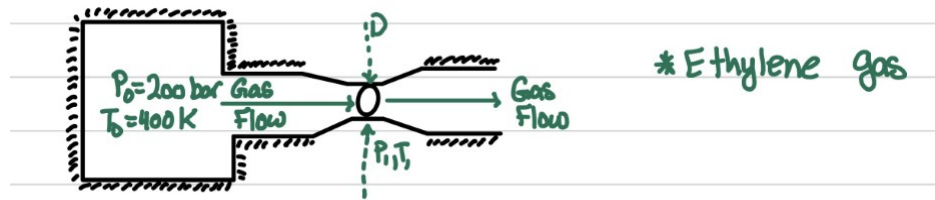


Figure 19: Hand Drawn Diagram

- Pressure drops, velocity increase until reaches speed of sound
- Volume = 20 m³
- Critical Condition
- Reversible and adiabatic
- Behaves as ideal gas
- KE $\neq$ 0

4.1 B.1. Initial mass of ethylene in the tank, kg:

MW : 28.0540 kg/kmol

$$\rho = \frac{P}{RT}$$

$$m = \rho V$$

$$R = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Conversion factor: 100 kJ = 1 bar · m³

$$\rho = \frac{(200 \text{ bar})}{\left(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}\right) (400 \text{ K})} = 0.0601395 \frac{\text{bar} \cdot \text{kmol}}{\text{kJ}} \times 100 \frac{\text{kJ}}{1 \text{ bar} \cdot \text{m}^3} = 6.013952 \frac{\text{kmol}}{\text{m}^3}$$

$$m = 6.013952 \frac{\text{kmol}}{\text{m}^3} \cdot 20 \text{ m}^3 \cdot 28.0540 \frac{\text{kg}}{\text{kmol}}$$

$$\boxed{m = 3,374.31 \text{ kg}}$$

4.2 B.2 The Temperature and Pressure at the narrow pipe section:

- $\dot{m} = 5000 \text{ kg/hr} \rightarrow$ Critical Condition
- corresponds to speed of sound
- Reversible and Adiabatic is an isentropic Process

$$-\Delta(mS) + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0$$

$\dot{Q} = 0$, adiabatic and $\dot{S}_{gen} = 0$, reversible so $\Delta S = 0$, isentropic

$$\text{we have : } \int_{T_1}^{T_2} C_p^{IG} \frac{dT}{T} = R \ln\left(\frac{P_2}{P_1}\right)$$

$$\text{and we know } \gamma = \frac{C_p}{C_v} \quad \text{and} \quad \frac{R}{C_p} = \frac{C_p - C_v}{C_p} = 1 - \frac{1}{\gamma}$$

$$\text{and } C_p^{IG} = \text{Constant} = C_p \quad \text{so,}$$

$$\begin{aligned}
C_p \ln\left(\frac{T_2}{T_1}\right) &= R \ln\left(\frac{P_2}{P_1}\right) \\
\ln\left(\frac{T_2}{T_1}\right) &= \frac{R}{C_p} \ln\left(\frac{P_2}{P_1}\right) \Rightarrow \ln\left(\frac{T_2}{T_1}\right) = \left[1 - \frac{1}{\gamma}\right] \ln\left(\frac{P_2}{P_1}\right) \\
\Rightarrow \ln\left(\frac{T_2}{T_1}\right) &= \ln\left[\left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}\right]
\end{aligned}$$

$$\frac{T_2}{T_1} = \left[\frac{P_2}{P_1}\right]^{1-\frac{1}{\gamma}} \quad (1)$$

Now,

$$\frac{dE_{CV}}{dt} = \dot{Q} - \dot{W} + \dot{m}_{in} \left[h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right] - \dot{m}_{out} \left[h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right]$$

It's a steady flow , adiabatic, no work, and PE=0 , so,

$$0 = 0 - 0 + \dot{m} \left[(h_{in} - h_{out}) + \left(\frac{V_{in}^2}{2} - \frac{V_{out}^2}{2} \right) \right]$$

and since there's no velocity at the beginning , $V_1 = 0$ m/s

$$0 = \dot{m} \left[(h_{in} - h_{out}) - \frac{V_{out}^2}{2} \right]$$

$$h_{in} = h_{out} + \frac{V_{out}^2}{2} \quad (2)$$

we know that $C_p = \left(\frac{\partial h}{\partial T} \right)_p \Rightarrow dh = C_p dT$

$$C_p T_{in} = C_p T_{out} + \frac{V_{out}^2}{2} \quad (3)$$

In the book we know the equation for the speed of sound is

$$v^2 = \frac{1}{M_w} \left(\gamma \frac{\partial P}{\partial \rho} \right)_T$$

and after derivations we get

$$v = \sqrt{\frac{\gamma R T}{M_w}}, \text{ where } \left(\frac{\partial P}{\partial \rho} \right)_T = R T, \text{ (pg.114)}$$

So since the velocity is equal to the speed of sound

$$\text{So } \Rightarrow C_p T_{in} = C_p T_{out} + \frac{\left(\sqrt{\frac{\gamma R T_{out}}{M_w}} \right)^2}{2}$$

$$\Rightarrow C_p T_{in} = C_p T_{out} + \frac{\gamma R T_{out}}{2 M_w}$$

$$\Rightarrow T_{in} = T_{out} + \frac{R}{C_p} \frac{\gamma T_{out}}{2 M_w}$$

$$T_{in} = T_{out} + \left(1 - \frac{1}{\gamma} \right) \frac{\gamma T_{out}}{2 M_w} \quad (4)$$

To find γ we will use the CHE Calculator to find C_p so :

$$\Rightarrow \text{CPIG(CompID, T)}$$

=> CPIG((Comp),(400-273.15)) , where Comp is ethylene and the T is temperature in °C

This gives $C_p = 54.2060 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

We Know $C_p - C_v = R$

$$C_v = C_p - R \Rightarrow C_v = 54.2060 - 8.314 = 45.8920 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

$$\gamma = \frac{C_p}{C_v} = \frac{54.2060}{45.8920} = 1.1812$$

Finding T_{out} first by Plugging everything in (4):

$$400K = T_{out} + \left(1 - \frac{1}{1.1812}\right) \frac{1.1812 T_{out}}{2 \left(28.0540 \frac{\text{kg}}{\text{kmol}}\right)} \cdot 28.0540 \frac{\text{kg}}{\text{kmol}}$$

$$400k = T_{out} + 0.0906T_{out}$$

$$400k = 1.0906T_{out}$$

$$\boxed{T_{out} = 366.771 \text{ K}}$$

Now finding P_{out} by Plugging into (1):

$$\frac{T_{out}}{T_{in}} = \left(\frac{P_{out}}{P_{in}} \right)^{1 - \frac{1}{\gamma}}$$

$$\frac{366.771}{400} = \left(\frac{P_{out}}{200} \right)^{1 - \frac{1}{1.1812}}$$

$$0.917 = \left(\frac{P_{out}}{200} \right)^{0.153}$$

$$0.568 = \frac{P_{out}}{200} \Rightarrow \boxed{P_{out} = 113.63 \text{ bar}}$$

4.3 B.3 The diameter of the narrow section , mm:

We know that :

$$\dot{m} = \rho \cdot v \cdot A \quad \text{where} \quad v = \text{Velocity} , \quad N_{out} = \text{Speed of sound} , \quad N_{out} = \sqrt{\frac{\gamma RT}{M_w}}$$

$$\rho = \text{density} \quad A = \text{Area} \quad \dot{m} = \text{mass flow rate}$$

$$\rho \text{ for ideal gas : } \rho = \frac{PM_w}{RT}$$

$$\rho = \frac{(113.63 \text{ bar})(28.0540 \text{ kg/kmol})}{(8.314 \text{ KJ/kmol.K})(366.771 \text{ K})} \cdot \left(\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \right) = 104.54 \text{ kg/m}^3$$

$$N_{out} = \sqrt{\frac{(1.1812)(8.314 \text{ KJ/kmol.k})(366.771 \text{ K})}{28.0540 \text{ kg/kmol}}} \cdot \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ KJ/Kg}} \right) = 358.316 \text{ m/s}$$

$$A = \frac{\dot{m}}{\rho \cdot N_{out}} = \frac{5000 \text{ kg/hr}}{(104.54 \text{ kg/m}^3)(358.316 \text{ m/s})} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 3.71 \times 10^{-5} \text{ m}^2$$

Now find the diameter :

$$d = 2\sqrt{\frac{A}{\pi}} = 2\sqrt{\frac{3.71 \times 10^{-5} \text{ m}^2}{\pi}} = 0.00687 \text{ m} \times \frac{10^3 \text{ mm}}{1 \text{ m}}$$

$$\boxed{6.87 \text{ mm}}$$

4.4 B.4 Mass of ethylene in tank when Critical Conditions are reached, kg:

This is Still ideal gas so,

$$P_1 V = m_1 R T_1 \quad \text{and} \quad P_2 V = m_2 R T_2$$

we also know:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{1-\frac{1}{\gamma}}$$

$$\frac{\frac{P_2 V}{m_2 R}}{\frac{P_1 V}{m_1 R}} = \left(\frac{P_2}{P_1} \right)^{1-\frac{1}{\gamma}}$$

$$\frac{P_2}{m_2} \cdot \frac{m_1}{P_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}} \quad \text{we need } \frac{m_2}{m_1} \text{ so take inverse :}$$

$$\frac{m_2}{P_2} \cdot \frac{P_1}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)}$$

$$\Rightarrow \frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)} \cdot \frac{P_2}{P_1} \quad * \quad -1 + \frac{1}{\gamma} + 1 = \frac{1}{\gamma}$$

$$\frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} \Rightarrow m_2 = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} m_1$$

$$\Rightarrow m_2 = \left(\frac{113.63}{200}\right)^{\frac{1}{1.1812}} (3,374.308) = \boxed{2,090.8069 \text{ kg}}$$

4.5 B.5 Mass of ethylene during the venting Process if $P_{gauge}=P_1=2300$

Psig , kg:

$$P_{sys} = P_{gauge} + P_{atm} \Rightarrow 2300 \text{ Psig} + 14.7 \text{ Psi} \Rightarrow 2314.7 \text{ Psia}$$

$$2314.7 \text{ Psia} \times \left(\frac{0.0689 \text{ bar}}{1 \text{ Psia}}\right) = 159.593 \text{ bar}$$

This is Still ideal gas so,

$$P_1 V = m_1 R T_1 \quad \text{and} \quad P_2 V = m_2 R T_2$$

we also know :

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}$$

$$\frac{\frac{P_2 V}{m_2 R}}{\frac{P_1 V}{m_1 R}} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}$$

$$\frac{P_2}{m_2} \cdot \frac{m_1}{P_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}} \quad \text{we need } \frac{m_2}{m_1} \text{ so take inverse :}$$

$$\frac{m_2}{P_2} \cdot \frac{P_1}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)}$$

$$\Rightarrow \frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)} \cdot \frac{P_2}{P_1} \quad * \quad -1 + \frac{1}{\gamma} + 1 = \frac{1}{\gamma}$$

$$\frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} \Rightarrow m_2 = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} m_1$$

Plug everything in:

$$m_2 = \left(\frac{159.593 \text{ bar}}{200 \text{ bar}}\right)^{\frac{1}{1.1812}} \cdot (3374.31 \text{ kg})$$

$$\boxed{m = 2,787.43 \text{ kg}}$$

5 Results Summary

Problem	Quantity	Reported Answer
A.1	β	0.00167124
A.2	ζ	0.00492405
A.3	H_{tot}	$-2.124659675 \frac{\text{KJ}}{\text{mol}}$
A.4	S_{tot}	$-75.55888391 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.5–6	$C_p - C_v$	$9.881915289 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.5	C_v	$23.32511018 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.6	C_p	$33.20702547 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.7	μ_{JT}	$-0.080431055 \frac{\text{K}}{\text{bar}}$
A.8	v	$796.8890813 \frac{\text{m}}{\text{s}}$
B.1	m (initial tank mass)	3,374.31 kg
B.2	T_{out}	366.771 K
B.2	P_{out}	113.63 bar
B.3	d	6.87 mm
B.4	m_2 (venting mass at $P_2 = 113.63$ bar)	2,090.8069 kg
B.5	m_2 (venting mass at $P_{\text{gauge}} = 2300$ Psig)	2,787.43 kg

6 Conclusion

In Problem A, the target state at 10°C and 830 bar was evaluated using interpolation and density-based thermodynamic identities. Akima interpolation (FAKIMA) was used to compute needed slopes/derivatives, construct integrands, and evaluate integrals for enthalpy, entropy, and C_v . From these results, additional properties were determined including C_p , the Joule–Thomson coefficient, and the speed of sound.

In Problem B, ethylene venting was analyzed under ideal-gas and isentropic assumptions to determine the initial tank mass, narrow-section temperature and pressure at the critical condition, the narrow-section diameter, and the mass remaining during the specified venting conditions. Overall, the assignment demonstrates how interpolation, derivative relations, and conservation laws are combined to compute thermodynamic and flow properties for engineering analysis.

7 References

References

- [1] OpenAI. *ChatGPT*. Used for formatting support, equation typing, and report editing. Accessed 2026.
- [2] Google. *Gemini*. Used for supplemental checking and assistance. Accessed 2026.
- [3] NIST. *NIST Chemistry WebBook*. National Institute of Standards and Technology. Used for thermophysical/property reference data. Accessed 2026.
- [4] CHE 301 Course Textbook Used for thermodynamic identities, derivative relations, and property definitions referenced throughout Problem A and Problem B. Spring 2026.